INNOVA JUNIOR COLLEGE
JC 2 MID YEAR EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

BIOLOGY

9744/04

Paper 4 Practical

17 August 2017

Confidential Instructions

2 hours 30 minutes

Great care should be taken to ensure that any confidential information, including the identity of material on microscope slides where appropriate, does not reach the candidates either directly or indirectly.

This document consists of **4** printed pages

Question 1

Preparation List

SN	Apparatus/ Reagents / Chemicals	Quantity per student
1	5 cm^3 syringes	1
2	1 cm^3 syringe	3
3	dropper	1
4	container / beaker labelled **For washing**	1
5	test-tubes	7
6	test-tube rack	1
7	50 cm^3 beaker	2
8	glass rod	1
9	white paper (as background for visualising colour)	1 sheet
10	stop watch	1
11	marker pen	1
12	safety goggles	1
13	distilled water, labelled **E**	5 cm^3
14	10% glucose, labelled **S1**	10 cm^3
15	2%, 4%, 6%, 8%, 10% glucose, labelled **G1, G2, G3, G4, G5**	5 cm^3 each
16	1M H_2SO_4, labelled **A**	10 cm^3
17	0.01% $KMNO_4$, labelled **P**	5 cm^3
18	distilled water, labelled **W**	10 cm^3

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Instructions for preparation

<u>To prepare **E**</u>
Provide distilled water, labelled **E**.

<u>To prepare **S1**</u>
10 % glucose solution, labelled **S1**.
This is prepared by dissolving 10 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water.

<u>To prepare **G1 – G5**</u>
2%, 4%, 6%, 8%, 10% glucose, labelled **G1 – G5**.
G1 is prepared by dissolving 2 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water.

G2 is prepared by dissolving 4 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water.

G3 is prepared by dissolving 6 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water

G4 is prepared by dissolving 8 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water.

G5 is prepared by dissolving 10 g of glucose in 75 cm^3 of distilled water and making up to 100 cm^3 with distilled water

<u>To prepare **A**</u>
1 mol dm^{-3} sulfuric acid, labelled **A**.
This is prepared from (98%) sulfuric acid by adding 55 cm^3 of the sulfuric acid to 500 cm^3 of distilled water and making up to 1 dm^3 with distilled water

This is an exothermic reaction, **add the acid to the water**.

<u>To prepare **P**</u>
0.01% potassium permanganate, labelled **P**.
This is prepared by dissolving 1.0 g of potassium permanganate in 100 cm^3 of distilled water. Then take 1 cm^3 of this solution and add to 99 cm^3 of distilled water.

Sulfuric acid is **harmful** and **corrosive;** potassium permanganate is **harmful** and should be disposed of with care to the environment.

It is advisable to wear safety glasses/goggles when handling these chemicals.

E, **S1** and **P** can be made up the day before the examination and stored in a refrigerator.
However, these must be at room temperature for the examination.

Question 2

Preparation List

SN	Apparatus/ Reagents / Chemicals	Quantity per student
1	light microscope with an eyepiece graticule, set up on **low power** objective lens	1
2	stage micrometer	1
3	Prepared slide of young root tip e.g. *Allium*, labelled **K1**	1